Lecture 1.1 – Intro to the Course

Specific Learning Objectives:

None

Agenda for today:

1. **Course Details**
2. **Why learn R?**
3. **About me**
4. **Install R and RStudio**

Student Office Hours

Instructor: Lindsay Waldrop, Ph.D.

Wed 1 am - 2:30 pm
Thu 10:30 am – 12 pm
& by appointment

What are office hours and why should you come?

- Office hours are for students! You are not bothering or interrupting us, this is time we set aside every week for my students!
- They are a great place to get clarification on content, ask questions, listen to other students ask questions, practice work, etc.
- They are also a great space for mentorship: career advice, info on graduate schools, exploring things that interest you, connecting with research, etc!

Course Navigation

- ***Where do I find course information?***

Github repository: https://github.com/CPSC-292/Fall2026-CourseInfo

- syllabus
- course learning objectives
- lecture notes
- course schedule
- course assignments
- sample code

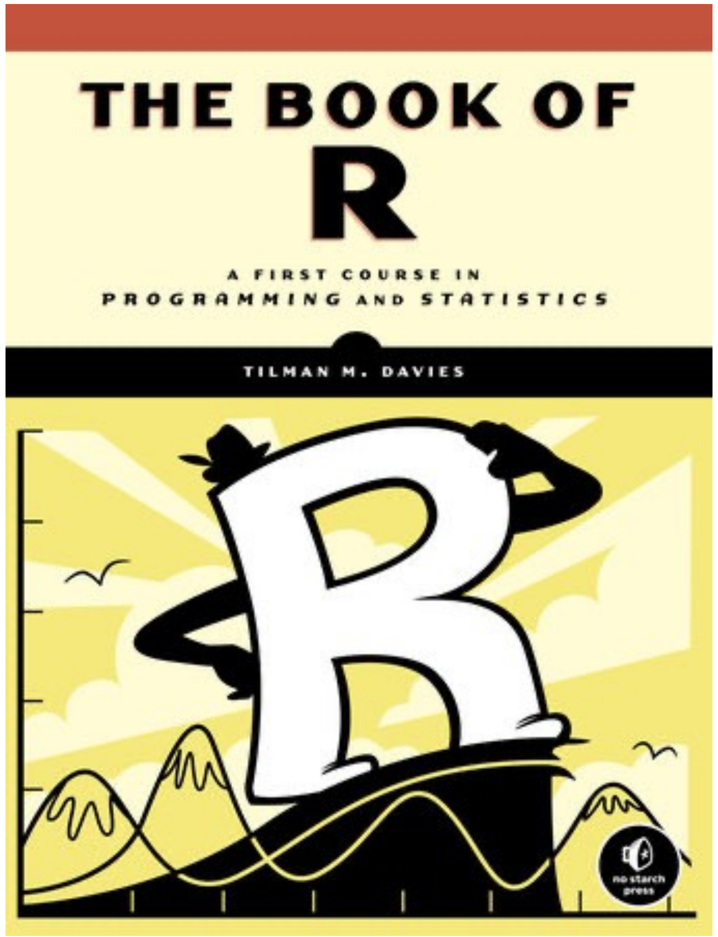
- ***Where do I find and turn in assignments and see my grades?***

Course Canvas site: https://canvas.chapman.edu/courses/88136

- ***What is the best way to communicate?***

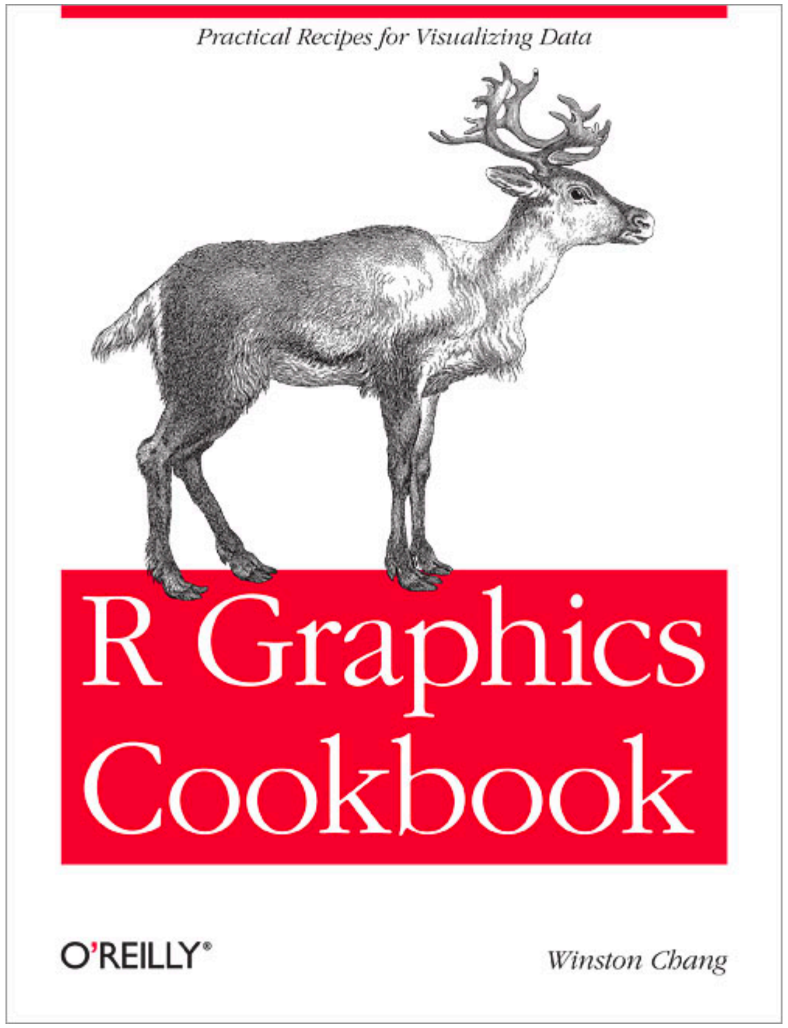
Slack Channel! Invite link in Canvas Module!

Course Materials



- **Required Text:** *The Book of R* by Tilman Davies.
  First Edition, No Starch Press. ISBN-13: 978-1-59327-651-5.
  Link to publisher website: https://nostarch.com/bookofr

Buy this book. There will be in-person quizzes on readings with no devices.



- **Suggested Text:** *R Graphics Cookbook* by Winston Chang.
  First Edition, O'Reilly Media. ISBN 9781491978603.
  Online at: https://r-graphics.org/

This is optional but I do recommend it!

Course Structure

- Course grade will be determined from a score of 100 translated to letter grades on a typical 10-point scale.

- Types of work and their weights for the course grade are:

1. **Assignments** (20%):

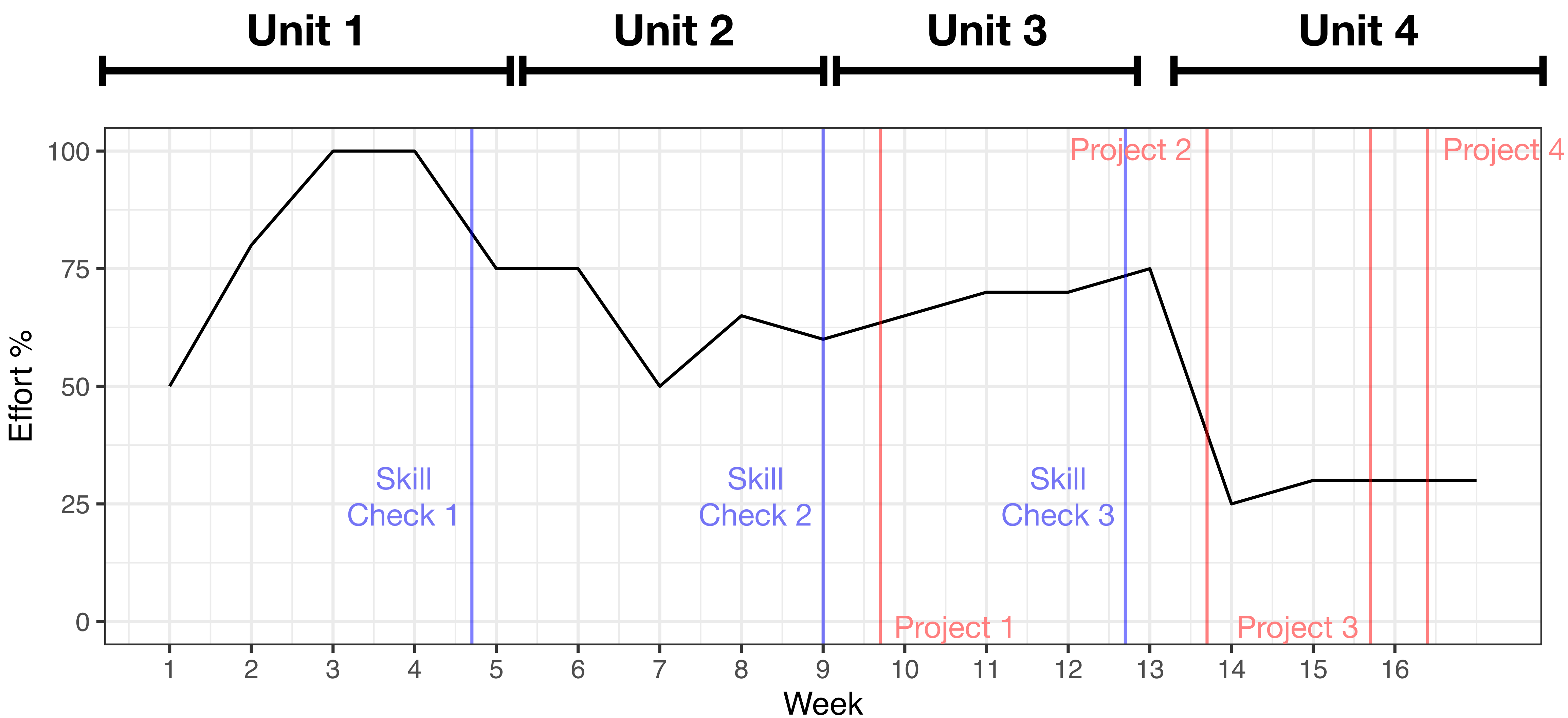
   - Out-of-class assignments (10%): coding practice with R and RStudio

   - Lecture assignments (10%): lecture videos and various pen-and-paper activities and practice (ex: group activities, worksheets, unannounced quizzes on lecture content or reading)

2. **Projects** (20%): out-of-class assignments designed on analysis of a chosen data set.

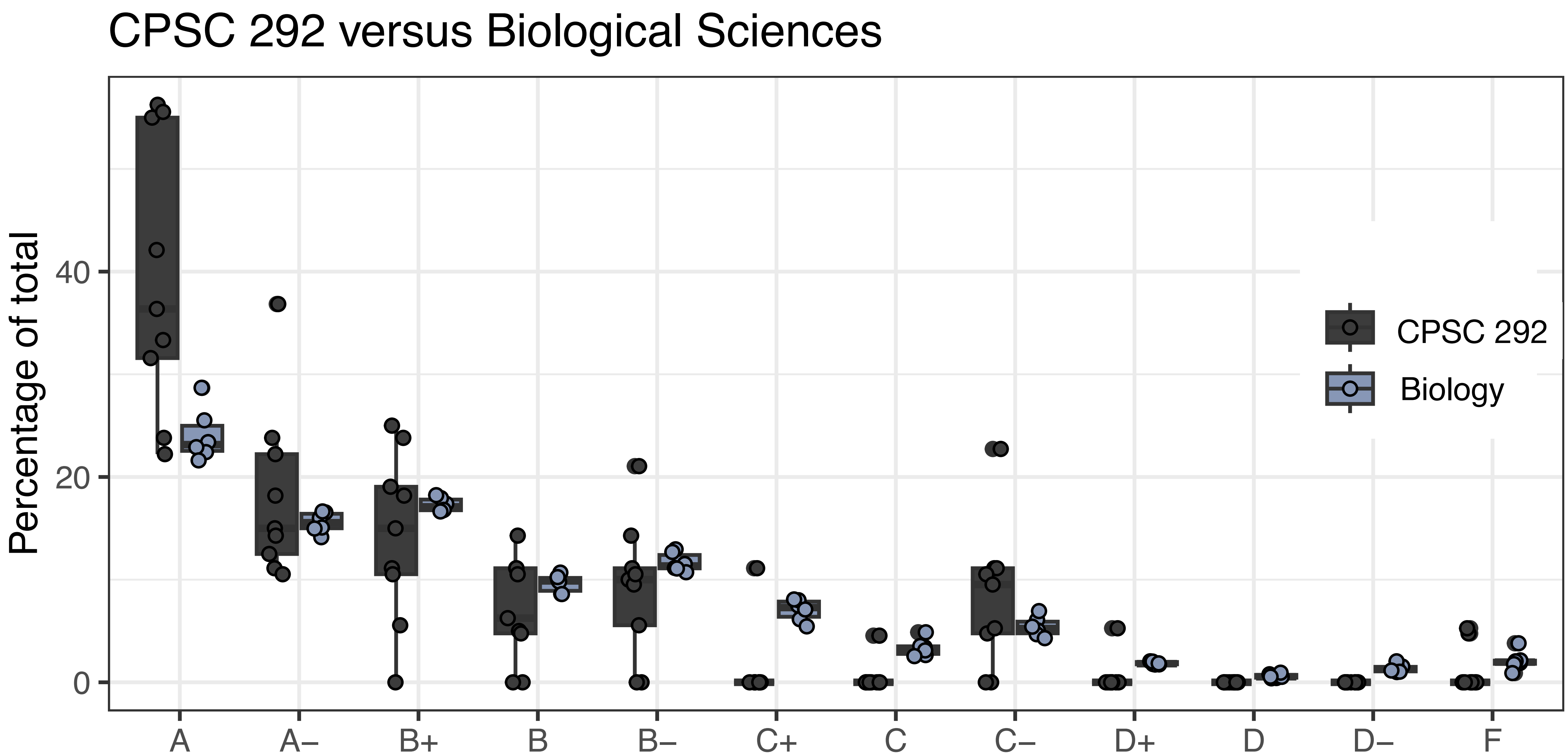
3. **Exams** (60%): in-class, pen-and-paper tests. Three chances to complete each problem on each exam. Three exams total.

A Word about the Amount of Work…



- This course's effort is front-loaded.
- Unit 1 moves quickly, contains half of all course assignments!
- **Critically important to keep up with assignments in Unit 1.**

…and what that means about grades.



More students earn A’s in my course than a typical biology course!

Other Course Policies

- Attendance will be managed using Qwickly. Only mark present if you are physically in class, do not help others if they are absent. Falsifying attendance records will result in failing the course.
- Final project policy: you MUST participate in the final project.
- Come prepared to learn and participate! There may be unannounced quizzes on lecture or reading material.
- No devices allowed in class by default. I will let you know if you can use computers or phones. Bring pen/pencil and paper every day.
- No smart glasses or other recording devices are allowed at any time.
- Group work is encouraged unless the assignment is an *individual evaluation*.
- Communication: Please Slack instead of email. Expect replies between 9 am and 5 pm during the regular work week.

Generative AI Use in this Course

- Gen AI is **limited use** in this course.

- **Unrestricted** generative AI use is allowed on any out-of-class assignment or project.

- AI use is **not allowed** on in-class assignments unless otherwise instructed.

- **No AI use is allowed on any exam.**

What if you need help?

- ***Please contact me!*** I want to help, whether it is situational, financial, or academic. I am prepared to be very flexible, including issuing course incompletes (which can be finished later).

- The Dean of Students can help connect you with services, no matter what type of problem you have!

- If you are struggling mentally, please talk to me or seek help through Student Psychological Counseling Services: https://www.chapman.edu/students/health-and-safety/psychological-counseling/

Course Learning Objectives

Main Learning Objectives:

1. Understand the basic structure and function of the *R* programming language.

2. Create visualizations and data analyses in the *R* programming language.

3. Independently perform basic data analysis and visualizations in a way that communicates ideas clearly.

Detailed learning objectives (and how they are assessed) are in the file CLO.pdf!

It’s true. Sorry :(

Why Learn *R*?

- Biology today involves lots and lots and lots of data!

- Most disciplines require skill in handling and analyzing data.

- *R* is a high-level yet powerful programming language that can assist with statistics, analysis, and visualization.

- *R* is free and open-source, making analyses replicable.

- *R* is flexible and has a huge community working on new stuff!

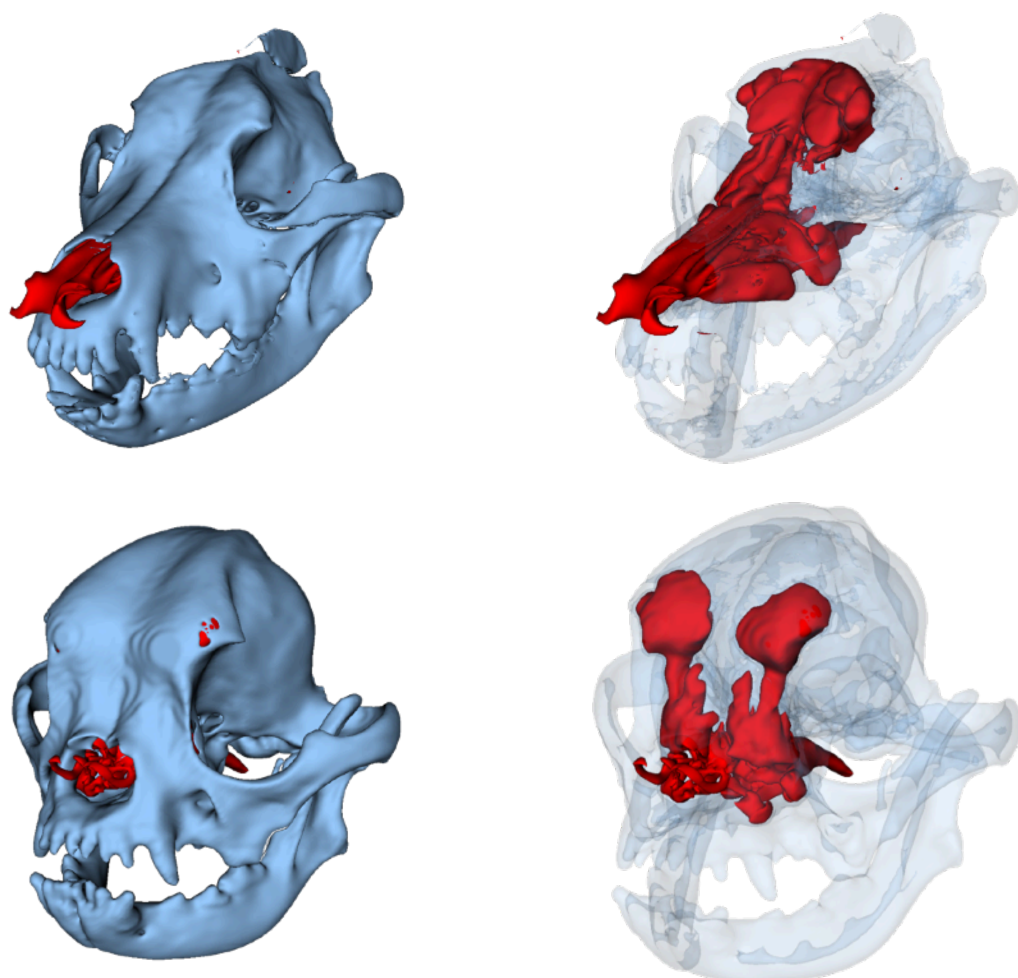
Why are you here?

About me

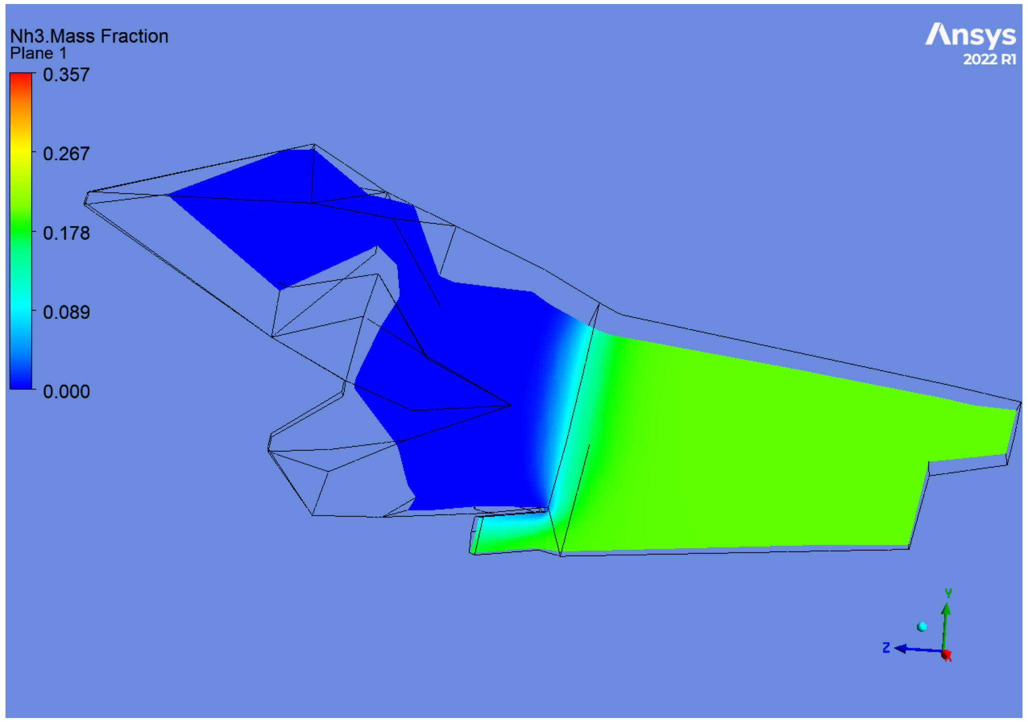
Lindsay Waldrop, Ph.D. (Biological Sciences) http://waldroplab.com

Exploring the evolution of biological fluid-structure interactions

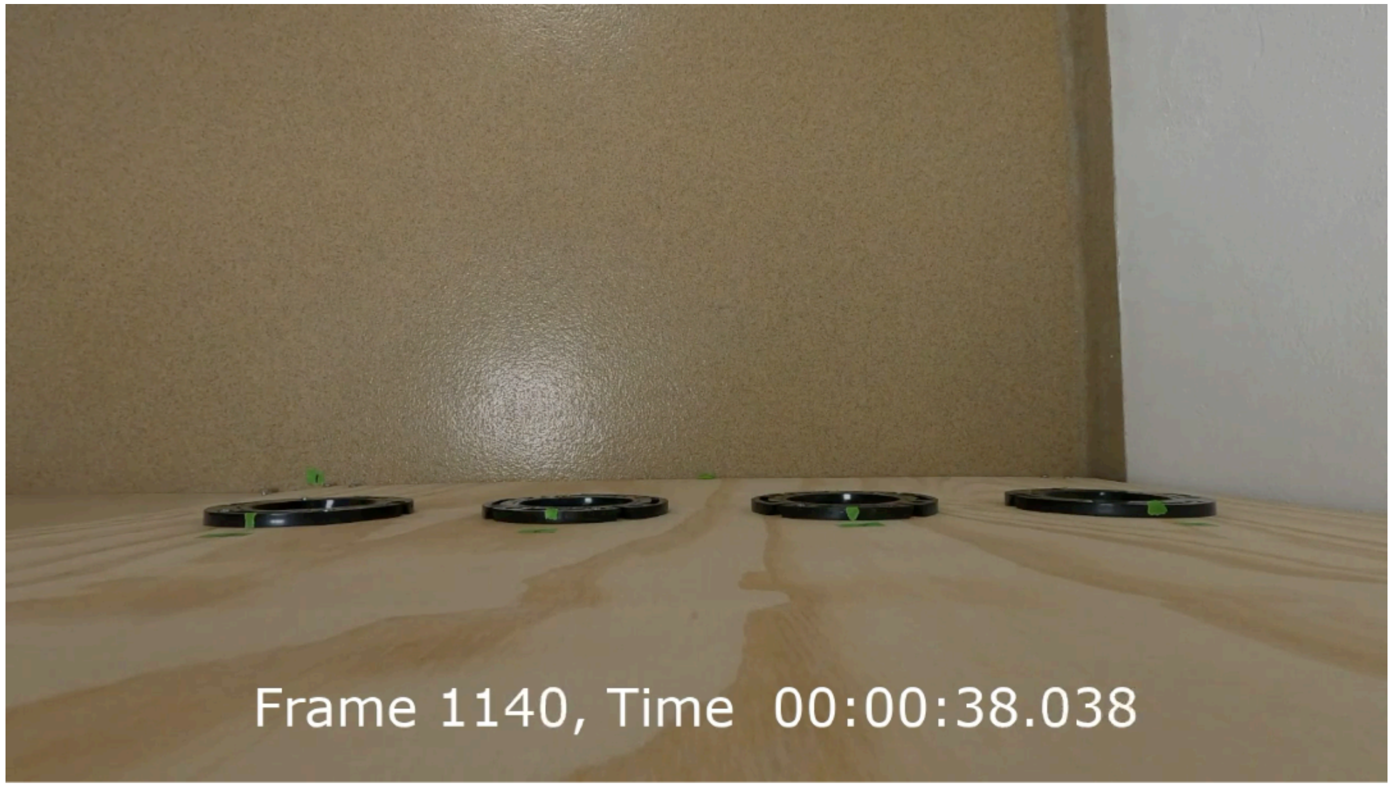
Morphometrics



Computational fluid dynamics



Experimental fluid dynamics



Spots open in my lab for dog projects!

Downloading *R* and RStudio

Download *R*:
https://www.r-project.org/

1. Go to https://cloud.r-project.org/
2. Select your operating system.
3. Select the latest release that is "notarized and signed."
4. Save and open the file, follow the instructions to install.

NOTE: be sure to select the correct chip type for Macs!

Download RStuido:
https://rstudio.com/products/rstudio/download/

1. Select the RStudio Desktop version.
2. Download, open, and follow instructions to install.
3. Open RStudio to get started!

Action Items

1. Have R and RStudio installed on your personal computer by Wed 9/3!

2. If you are having trouble with a laptop, please let me know!